



an end user or a content generator. Think of Palladium as a policing system whose actions are guided by laws set by a third party, akin to the judiciary.

Think secure

Palladium is meant to cultivate a computing environment, termed as 'trusted space', which a user can rely on for security and privacy. At the heart of the system lies a secure cryptographic co-processor, that is, a tamper-resistant embedded smart card, which contains unique keys and a variety of cipher implementations. It thus acts as a unique identifier of a particular system and is the base on which Palladium's security measures are implemented. Additional changes made to the CPU (in terms of extensions, similar to 3DNow!) will allow the central processor to run in a special mode. Finally, to enable secure communication over a motherboard's data buses, the chip too will have to be modified.

Built upon this hardware foundation and extending its functionality is a tiny piece of software termed the Nexus. It represents a change in the kernel of the operating system and takes the form of an independent micro-kernel. The Nexus is run in an ultra-privileged mode of the CPU where certain areas of memory are strictly reserved for it and are restricted to other,

potentially malicious, pieces of code. This micro-kernel dictates the rudimentary security policies that the rest of the system must follow. Under Palladium, a Nexus can be changed, thus changing the security parameters and guidelines. Under Palladium, these rudimentary policies can be changed by using a different Nexus.

The Nexus is the manager of trusted memory and an arbitrator of access to the cryptographic co-processor. When a Palladium PC is booted for the first time, an inventory of the system hardware is taken and authenticated. If a part of the hard-

cations do not trespass into restricted memory or file systems. Thanks to the way a CPU works, it is not possible for software to modify the Nexus or the policies laid down by it.

The Nexus is thus protected by hardware, while it protects the software that forms the applications and data. The Nexus maintains a low profile, in a general stance, entities known as Trusted Agents (TA) do the legwork of interacting with the user. A TA can either be a Palladium-enabled application or an add-on to an existing program, imparting Palladium awareness to it.

Although existing applications will run on a Palladium system, they will not be able to use the security features without TAs. To better comprehend the relation between applications and TAs, consider the analogy of a language. If the language defines the rules of grammar, the TA uses these rules to construct sentences as a means of communicating securely with other systems and users.

Think trust

The Nexus and TAs together provide a secure data storage, encryption services, applications and facilities to enable hardware and software to authenticate themselves and each other. Central to the Palladium concept are 'Spheres of Trust' wherein a user can not only define a program to trust but can also ask it to store only a certain set of data. Its like trusting your doctor with very sensitive health information. You are unlikely, however, to share with him the particulars of your bank account—details that your broker is aware of. The broker, in turn, is in the know about the details of your health. The trust you lend to your physician does not extend into your financial life, just as a broker's 'sphere of trust' does not encompass information on your health.

Similarly, Palladium divides usage into separate spheres or vaults. A trusted application is granted access to either a section or an entire vault. The user thus controls what is revealed to whom. This policy is akin to the one that can be set within Internet Explorer and it can be used to grant you a secure environment while working on a low or zero-security space while playing games or movies—all on the same machine and in the same user session.



John Manfredelli
GENERAL MANAGER,
MICROSOFT PALLADIUM
UNIT

“Palladium is a new hardware and software architecture that will run applications in a memory space that is highly resistant to tampering and interference”

